1 Mask

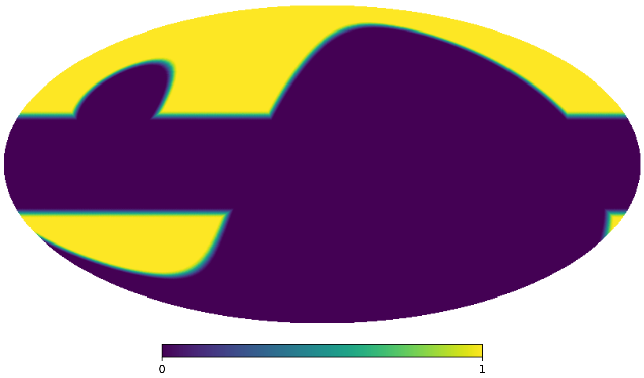

Figure 1: Analysis mask combining the QUIJOTE low-declination survey mask with a 20° cut in galactic latitude.

2 Data

Our analysis employs two complementary datasets: simulated sky maps generated using the `Python` Sky Model (`PySM`) library, which incorporates all major Galactic foreground components; and observational data from the QUIJOTE, WMAP, and Planck experiments. All simulated maps are convolved with the appropriate beam window functions for each respective experiment to ensure realistic instrumental effects.

Before fitting the observed power spectra, we subtract the CMB contribution using the best-fit theoretical ΛCDM spectrum from Planck 2018 (Planck Collaboration, 2020). This spectrum is obtained from the full Plik TTTEEE likelihood combined with low-multipole (lowE) and gravitational lensing constraints. For polarization modes, we subtract both C_ℓ^{EE} and C_ℓ^{BB}, where the latter includes only the contribution from B-modes generated by gravitational lensing, since primordial B-modes of inflationary origin have not yet been detected. This CMB subtraction is essential to isolate the Galactic foreground emission (synchrotron and thermal dust) that constitutes the target of this analysis.

3 Error estimation

Power spectrum uncertainties are estimated through Monte Carlo simulations for both datasets. We compute the standard deviation from 100 realizations each of sky+noise and pure noise maps. The noise characteristics vary by experiment: QUIJOTE and Planck employ the official noise simulations provided by their respective collaborations, incorporating both white noise and correlated $1/f$ noise components. For WMAP, we model the noise as purely white, following standard practice for this dataset.

4 MCMC fitting

The total angular power spectrum between frequencies ν_i and ν_j is modeled as

$$C_\ell(\nu_i, \nu_j) = C_\ell^{\text{sync}}(\nu_i, \nu_j) + C_\ell^{\text{dust}}(\nu_i, \nu_j) + C_\ell^{\text{sd}}(\nu_i, \nu_j), \tag{1}$$

where the synchrotron, dust, and synchrotron-dust cross-correlation terms are

$$C_\ell^{\text{sync}}(\nu_i, \nu_j) = A_s \left(\frac{\ell}{\ell_{\text{ref}}}\right)^{\alpha_s} \left(\frac{\nu_i}{\nu_{\text{ref}}}\right)^{\beta_s} \left(\frac{\nu_j}{\nu_{\text{ref}}}\right)^{\beta_s}, \tag{2}$$

$$C_\ell^{\text{dust}}(\nu_i, \nu_j) = A_d \left(\frac{\ell}{\ell_{\text{ref}}}\right)^{\alpha_d} S_d(\nu_i) S_d(\nu_j), \tag{3}$$

$$\begin{aligned} C_\ell^{\text{sd}}(\nu_i, \nu_j) = \rho \sqrt{A_s A_d} \Big[& S_s(\nu_i) S_d(\nu_j) \\ & + S_s(\nu_j) S_d(\nu_i) \Big] \left(\frac{\ell}{\ell_{\text{ref}}}\right)^{(\alpha_s + \alpha_d)/2}, \end{aligned} \tag{4}$$

with $S_s(\nu) = (\nu/\nu_{\text{ref}})^{\beta_s}$ and the modified blackbody scaling in Rayleigh-Jeans units

$$S_d(\nu) = \frac{B_\nu(\nu, T_d)}{B_\nu(\nu_0, T_d)} \left(\frac{\nu}{\nu_0}\right)^{\beta_d}, \tag{5}$$

where B_ν is the Planck function. Here, A_s and A_d are the amplitudes of the synchrotron and dust components at the reference multipole $\ell_{\text{ref}} = 80$ and frequencies $\nu_{\text{ref}} = 11.1$ GHz and $\nu_0 = 353.0$ GHz, respectively; α_s and α_d are the multipole spectral indices; β_s and β_d are the frequency spectral indices; T_d is the dust temperature (fixed at $T_d = 19.6$ K); and ρ is the synchrotron-dust correlation coefficient.

4.1 Data Selection and Frequency Coverage

We perform a Markov Chain Monte Carlo (MCMC) fitting procedure using data from QUIJOTE 11 GHz; WMAP 23, 33, 41, 61, and 94 GHz; and Planck 30, 44, 70, 100, 143, 217, and 353 GHz, covering the multipole range $30 < \ell < 200$. In this study, we use both auto- and cross-spectra between frequency bands. This yields 91 independent spectra from the 13 frequency bands.

4.2 Bayesian Inference

We adopt a Bayesian framework to constrain the model parameters $\boldsymbol{\theta} = \{A_s, \alpha_s, \beta_s, A_d, \alpha_d, \beta_d, \rho\}$. The posterior distribution is given by

$$p(\boldsymbol{\theta} \mid \mathbf{D}) \propto \mathcal{L}(\mathbf{D} \mid \boldsymbol{\theta})\, p(\boldsymbol{\theta}), \tag{6}$$

where $\mathcal{L}(\mathbf{D} \mid \boldsymbol{\theta})$ is the likelihood and $p(\boldsymbol{\theta})$ is the prior. The likelihood is constructed from the observed power spectra $\hat{C}_\ell(\nu_i, \nu_j)$ and their uncertainties:

$$\ln \mathcal{L} = -\frac{1}{2} \sum_{i,j,\ell} \frac{[\hat{C}_\ell(\nu_i, \nu_j) - C_\ell(\nu_i, \nu_j; \boldsymbol{\theta})]^2}{\sigma_\ell^2(\nu_i, \nu_j)}. \tag{7}$$

For the power-law model fitting, we adopt uniform (flat) priors for all model parameters within physically motivated bounds:

$$A_s \in [0, \infty), \tag{8}$$
$$\alpha_s \in [-6, 0], \tag{9}$$
$$\beta_s \in [-6, 0], \tag{10}$$
$$A_d \in [0, \infty), \tag{11}$$
$$\alpha_d \in [-6, 0], \tag{12}$$
$$\beta_d \in [0, 6], \tag{13}$$
$$\rho \in [-1, 1], \tag{14}$$

For the bin-to-bin analysis, to regularize the fit and incorporate external information from the literature, we impose a Gaussian prior on the synchrotron frequency spectral index:

$$\beta_s \sim \mathcal{N}(-3.1, 0.30^2), \tag{15}$$

while maintaining uniform priors for the remaining parameters within the bounds specified above.

4.3 Fitting methods

We apply three complementary approaches to the angular power spectra computed from the maps:

Power-law fitting, $\Delta\ell = 10$ We compute the spectra with a bin width $\Delta\ell = 10$ and fit a power-law model for the multipole dependence of the primary foreground components (synchrotron and dust). This fit estimates the amplitudes (A_s, A_d), the frequency spectral indices (β_s, β_d), the multipole spectral indices (α_s, α_d), and the correlation coefficient (ρ) between components.

Power-law fitting, $\Delta\ell = 20$ We repeat the power-law fit using a wider binning, $\Delta\ell = 20$, to test the stability of the fitted parameters with respect to the choice of bin size. Wider bins reduce per-bin noise at the cost of multipole resolution.

Bin-to-bin analysis, $\Delta\ell = 20$ In this approach each multipole bin is fitted independently with $\Delta\ell = 20$, without assuming a global power-law dependence on ℓ. For each bin we only fit the amplitudes (A_s, A_d), the frequency spectral indices (β_s, β_d), and the correlation (ρ); the multipole spectral indices (α_s, α_d) are not constrained in this analysis. This method allows us to identify scale-dependent behaviour and local deviations from the power-law assumption.

5 Results

In the following subsections we first present the results obtained on simulated data and then the results for observational data, applying the three procedures described previously (Section 4.3).

5.1 Observational data

We show results for the same set of fitting approaches used on the simulations to allow direct comparison.

5.1.1 Power-law fit ($\Delta\ell = 20$)

The power-law fit with wider bins ($\Delta\ell = 20$) is shown in Figure 2 (EE) and Figure 3 (BB).

5.1.2 Bin-to-bin analysis ($\Delta\ell = 20$)

The bin-to-bin analysis for observational data fits each multipole bin independently; results are listed in Table 1 and illustrated in Figure 4.

References

Planck Collaboration (2020). Planck 2018 results. VI. Cosmological parameters. *A&A*, 641:A6.

Table 1: Bin-to-bin fit results. EE mode (top) and BB mode (bottom).

ℓ range	$\ell_{\rm eff}$	$A^{\rm EE}_{\rm sync}$ [μK^2]	$\beta^{\rm EE}_{\rm sync}$	$A^{\rm EE}_{\rm dust}$ [10^{-3} μK^2]	$\beta^{\rm EE}_{\rm dust}$	$\rho^{\rm EE}$
20–39	29.0	26.56 $\pm$ 0.82	-3.12 $\pm$ 0.02	22.18 $\pm$ 0.26	1.45 $\pm$ 0.01	0.003 $\pm$ 0.010
40–59	49.0	3.42 $\pm$ 0.25	-3.13 $\pm$ 0.05	4.45 $\pm$ 0.07	1.48 $\pm$ 0.02	0.074 $\pm$ 0.015
60–79	69.0	1.24 $\pm$ 0.18	-3.21 $\pm$ 0.33	1.74 $\pm$ 0.04	1.41 $\pm$ 0.11	-0.094 $\pm$ 0.024
80–99	89.0	0.60 $\pm$ 0.11	-3.19 $\pm$ 0.33	1.26 $\pm$ 0.07	1.45 $\pm$ 0.13	0.049 $\pm$ 0.025
100–119	109.0	0.18 $\pm$ 0.06	-3.02 $\pm$ 0.22	0.88 $\pm$ 0.02	1.53 $\pm$ 0.03	0.067 $\pm$ 0.099
120–139	129.0	0.10 $\pm$ 0.05	-2.88 $\pm$ 0.38	0.46 $\pm$ 0.02	1.29 $\pm$ 0.09	0.182 $\pm$ 0.088
140–159	149.0	0.14 $\pm$ 0.05	-2.82 $\pm$ 0.25	0.35 $\pm$ 0.01	1.42 $\pm$ 0.04	0.097 $\pm$ 0.060
160–179	169.0	0.15 $\pm$ 0.07	-3.15 $\pm$ 0.42	0.29 $\pm$ 0.01	1.51 $\pm$ 0.14	0.048 $\pm$ 0.085
180–199	189.0	0.05 $\pm$ 0.05	-3.09 $\pm$ 0.29	0.20 $\pm$ 0.01	1.58 $\pm$ 0.07	0.127 $\pm$ 0.214
ℓ range	$\ell_{\rm eff}$	$A^{\rm BB}_{\rm sync}$ [μK^2]	$\beta^{\rm BB}_{\rm sync}$	$A^{\rm BB}_{\rm dust}$ [10^{-3} μK^2]	$\beta^{\rm BB}_{\rm dust}$	$\rho^{\rm BB}$
20–39	29.0	3.63 $\pm$ 0.38	-3.05 $\pm$ 0.07	8.95 $\pm$ 0.13	1.48 $\pm$ 0.02	0.182 $\pm$ 0.021
40–59	49.0	0.87 $\pm$ 0.20	-3.17 $\pm$ 0.34	3.15 $\pm$ 0.12	1.45 $\pm$ 0.09	-0.003 $\pm$ 0.100
60–79	69.0	0.11 $\pm$ 0.07	-3.10 $\pm$ 0.41	1.45 $\pm$ 0.07	1.57 $\pm$ 0.06	0.140 $\pm$ 0.110
80–99	89.0	0.05 $\pm$ 0.04	-3.13 $\pm$ 0.42	0.82 $\pm$ 0.03	1.54 $\pm$ 0.08	-0.017 $\pm$ 0.166
100–119	109.0	0.05 $\pm$ 0.03	-2.93 $\pm$ 0.32	0.50 $\pm$ 0.01	1.49 $\pm$ 0.03	0.055 $\pm$ 0.142
120–139	129.0	0.06 $\pm$ 0.05	-3.09 $\pm$ 0.27	0.30 $\pm$ 0.01	1.55 $\pm$ 0.04	0.124 $\pm$ 0.147
140–159	149.0	0.02 $\pm$ 0.03	-3.11 $\pm$ 0.30	0.25 $\pm$ 0.01	1.52 $\pm$ 0.04	0.053 $\pm$ 0.235
160–179	169.0	0.19 $\pm$ 0.08	-3.25 $\pm$ 0.26	0.19 $\pm$ 0.01	1.48 $\pm$ 0.05	-0.086 $\pm$ 0.071
180–199	189.0	0.14 $\pm$ 0.06	-2.79 $\pm$ 0.24	0.15 $\pm$ 0.01	1.49 $\pm$ 0.06	-0.075 $\pm$ 0.073

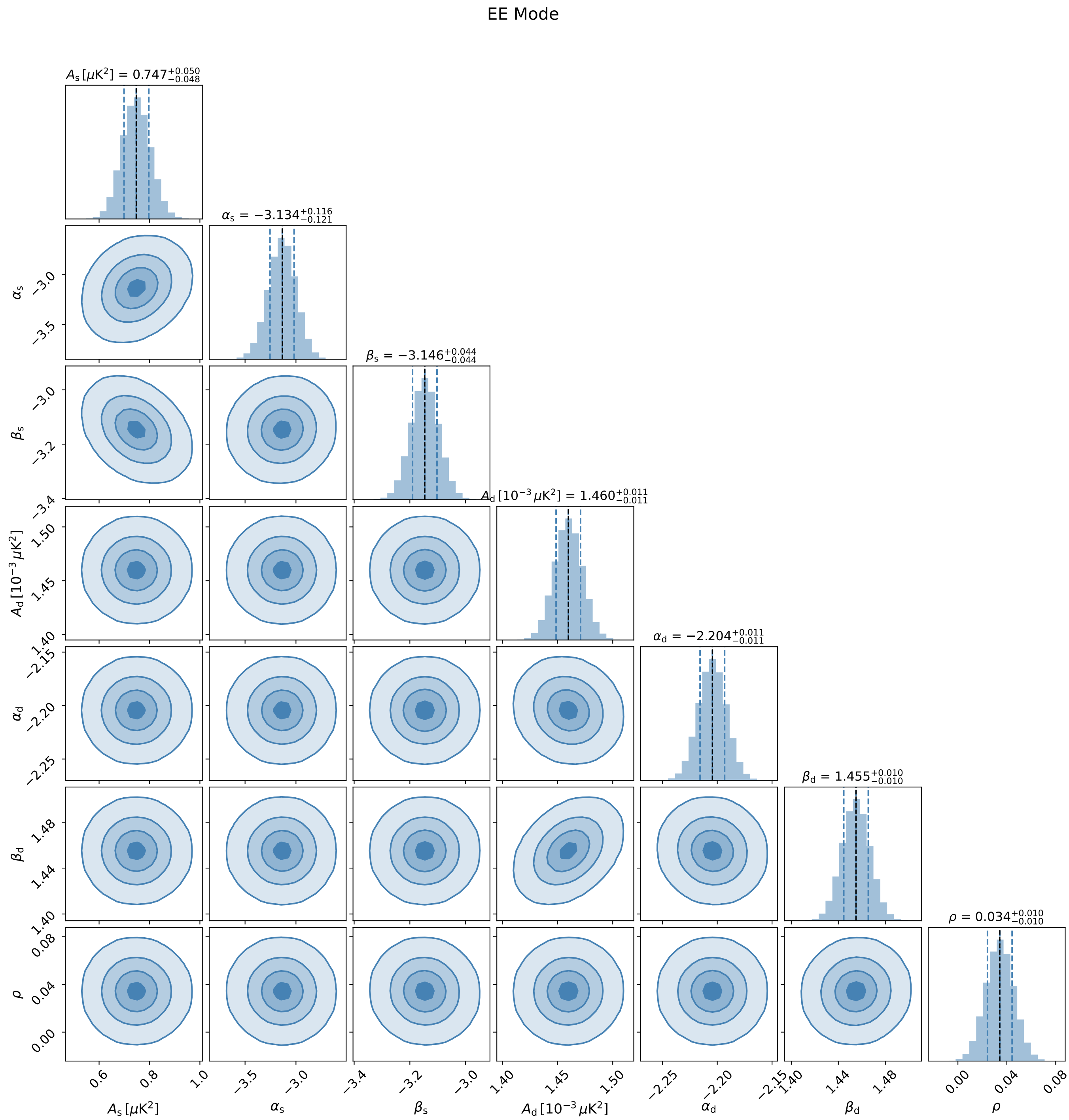

Figure 2: Corner plot showing posterior distributions for synchrotron and dust model parameters from MCMC fitting of observational EE mode data with bin width $\Delta\ell = 20$.

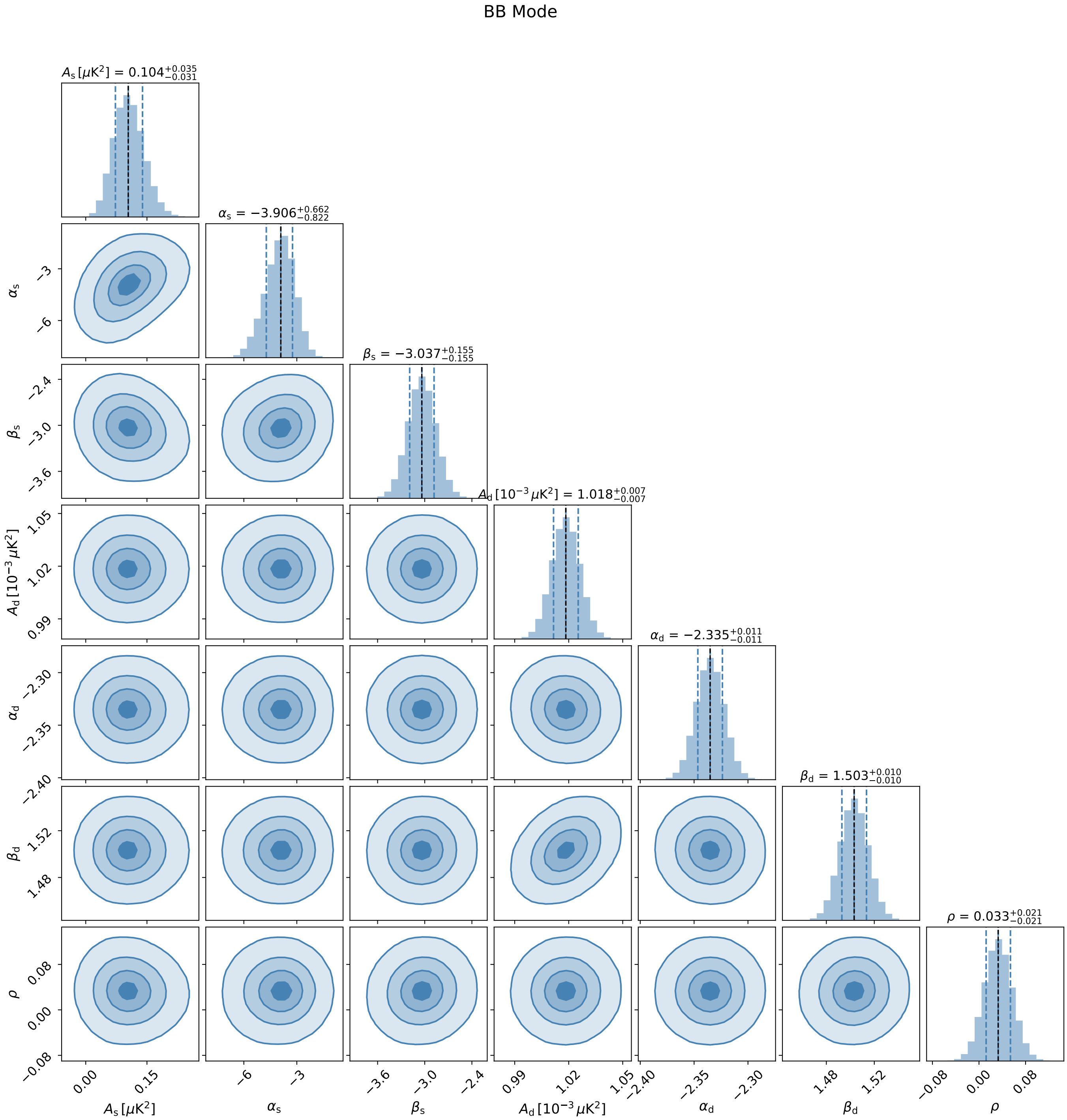

Figure 3: Corner plot showing posterior distributions for synchrotron and dust model parameters from MCMC fitting of observational BB mode data with bin width $\Delta\ell = 20$.

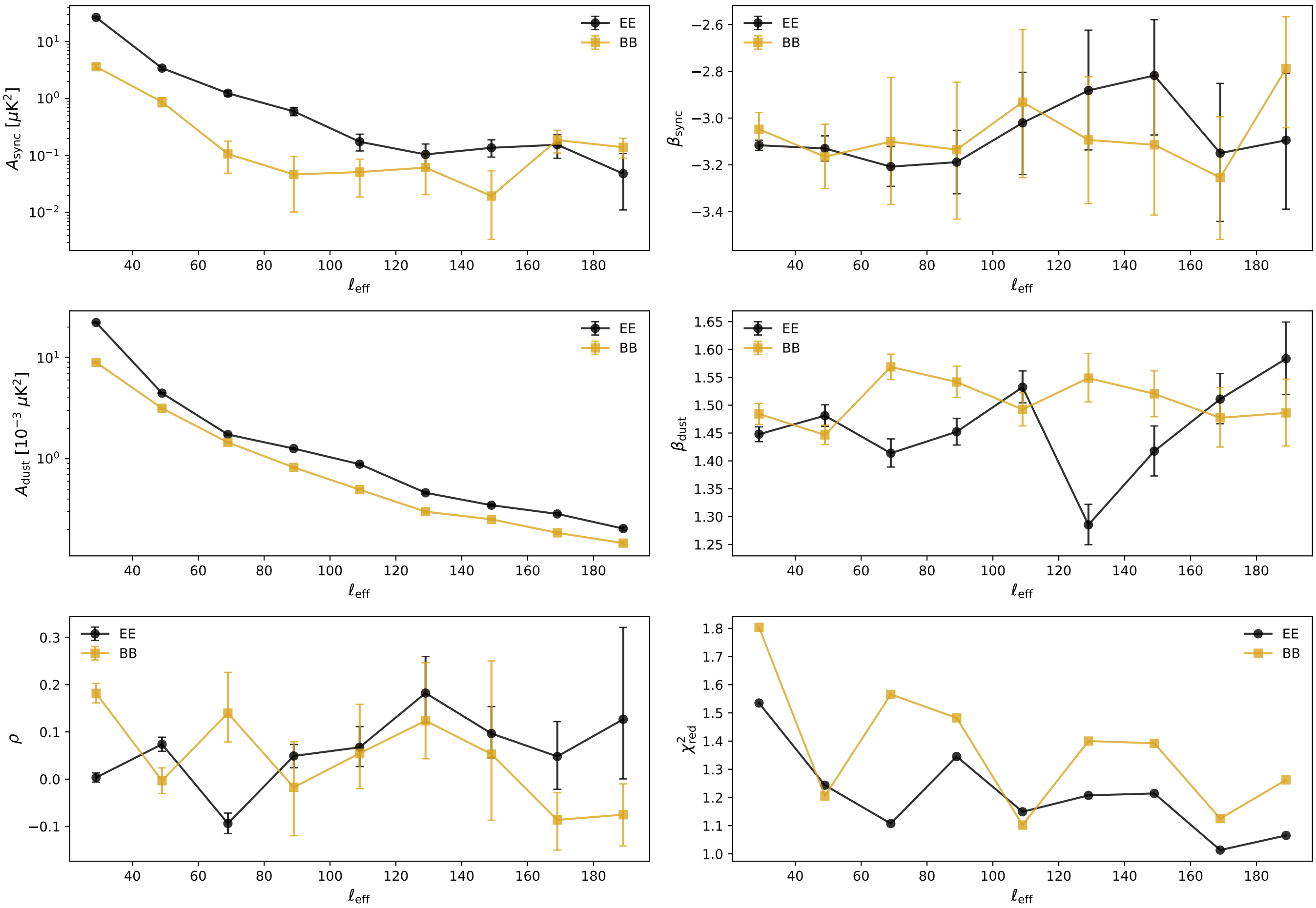

Figure 4: Bin-to-bin analysis results (observational) showing the evolution of fitted synchrotron and dust parameters as a function of multipole ℓ for both EE (top) and BB (bottom) modes.